Lecture 20 – Resistivity, Conductivity & Microscopic Ohm's Law

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Contents

1 Practice with $R = \rho \frac{L}{A}$ **2**

1.1 Question 30 2

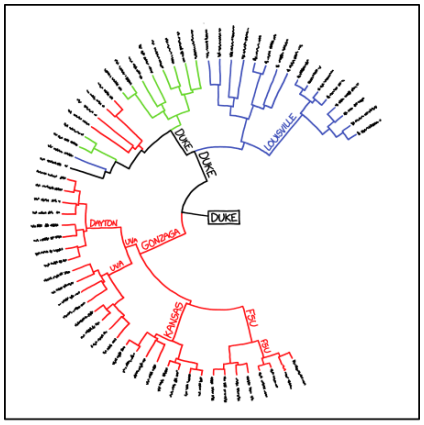
1.2 Question 71 2

1.3 Resistivity of Muscle and Bone 2

2 Ohm's Law in Detail **4**

3 Microscopic Ohm's Law **4**

3.1 Question 21 5



I WAS KICKED OFF THE BIOLOGY PROJECT AFTER I SECRETLY REPLACED ALL THE PHYLOGENETIC TREES IN OUR NEW PAPER WITH MARCH MADNESS BRACKETS.

1 Practice with $R = \rho\frac{L}{A}$

1.1 Question 30

How long must a 0.6mm diameter aluminium wire be to have a 0.5A current when connected to a AA battery?

$$R = \frac{V}{I} = \rho\frac{L}{A}$$

$$L = \frac{VA}{I\rho} = \frac{1.5 * \pi(0.3 \times 10^{-3})^2}{0.5 * 2.8 \times 10^{-8}}$$

1.2 Question 71

A long round wire has resistance R_0. What is its resistance if you stretch it to double it's length?

Volume, or $V = AL$, stays constant the whole time. Length and area, however, scale.

$$R = \rho\frac{L}{A}$$

and,

$$A = \frac{V}{L}$$

$$R = \rho\frac{L}{\frac{V}{L}}$$

$$R = \rho\frac{L^2}{V}$$

Now, since we only have one variable, we can safely say that if L doubles, R quadruples.

1.3 Resistivity of Muscle and Bone

Suppose one were to measure a current of 0.58mA when a 0.60V potential difference is applied between my feet. What percentage fat are my legs?

$$\rho_{fat} = 25\Omega m$$

$$\rho_{musc} = 13\Omega m$$

$$\rho_{bone} = \infty$$

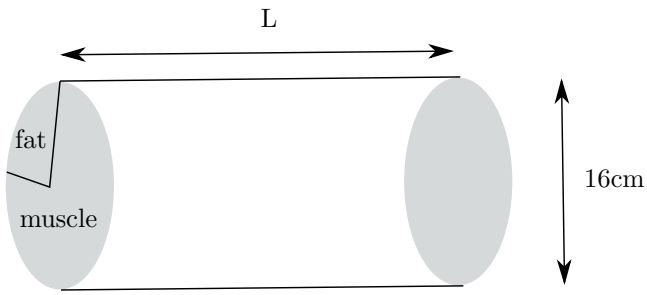


Figure 1: Totally accurate leg diagram

$$A = \pi r^2, A_{fat} = (1-x)A, A_{musc} = xA$$

$$R_{fat} = \frac{\rho_{fat}}{A_{fat}} = \frac{\rho_{fat}L}{(1-x)A}$$

$$R_{musc} = \frac{\rho_{musc}L}{xA}$$

$$I_{tot} = I_{fat} + I_{musc} = \frac{V}{R_{fat}} + \frac{V}{R_{musc}}$$

$$= \frac{V(1-x)A}{\rho_{tot}L} + \frac{VxA}{\rho_{musc}L}$$

$$\frac{LI_{tot}}{VA} = \frac{1-x}{\rho_{fat}} + \frac{x}{\rho_{musc}}$$

$$= \frac{1}{\rho_{fat}} - \frac{x}{\rho_{fat}} + \frac{x}{\rho_{musc}}$$

Finally,

$$\frac{LI_{tot}}{VA} - \frac{1}{\rho_{fat}} = x(\frac{1}{\rho_{musc} - \frac{1}{\rho_{fat}}})$$

$$x = \frac{\frac{LI_{tot}}{VA} - \frac{1}{\rho_{fat}}}{\frac{1}{\rho_{musc}} - \frac{1}{\rho_{fat}}}$$

$$x = 87$$

The answer is 87% muscle, 13% fat.

2 Ohm's Law in Detail

'Perfect' resistors have a very linear slope. 'Realistic' resistors, however, are more of a decreasing exponential function.

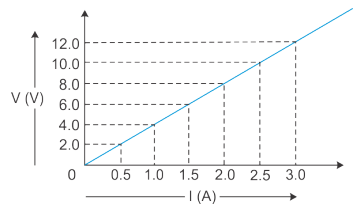


Figure 2: Perfect resistor graph

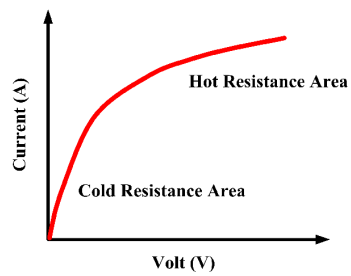


Figure 3: Realistic resistor graph

3 Microscopic Ohm's Law

Ohm's law behaves somewhat differently on the microscopic scale.

$$V = IR$$

In which:

- $V = E = \frac{V}{L}$
- $I = J := \frac{I}{A}$, or current density (current per cross-sectional unit area)
  - Alternate representation: $\vec{J} = \sigma \vec{E}$, where sigma is the sigma conductivity $\sigma = \frac{1}{\rho}$
- $R = \rho \frac{L}{A}$

3.1 Question 21

A $0.0075\frac{V}{m}$ electric field creates a 3.9mA current in a 1mm diameter wire. What is the wire made of?

$$J = \sigma E$$

$$\sigma = \frac{I}{EA}$$

$$\frac{I}{A} = \frac{I}{E\pi(\frac{d}{2})^2} = 660000\Omega^{-1}m^{-1}$$

Material is nichrome